CHAPTER FOUR

RESULT

4.1 Socio-Demographic Characteristics of Respondents

This section describes the demographic profile of the respondents, focusing on sex, marital status, and educational level. A total of 400 respondents participated in the study. Understanding the socio-demographic background is essential in interpreting variations in knowledge, attitudes, and practices related to open defecation. The data presented in Tables 4.1 and 4.2 provide insights into the composition of the study population and how these characteristics may influence sanitation-related behaviors.

Table 4.1 presents the sex distribution of respondents. The results show that 56.8% of respondents were male, while 43.3% were female. This slightly higher proportion of male participants could influence the findings, as gender may affect perceptions, responsibilities, and practices related to sanitation and open defecation.

Table 4.2 shows the educational level of respondents. The majority (40.5%) had secondary education, followed by 34.8% with primary education, 17.8% with tertiary education, and only 7.0% with no formal education. This indicates that most respondents possessed basic literacy, which may influence their understanding of sanitation-related risks and adoption of healthy hygiene practices.

Table 4.1: Sex of Respondents

Sex

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	173	43.3	43.3	43.3
	Male	227	56.8	56.8	100.0
	Total	400	100.0	100.0	

Table 4.2: Educational Level of Respondents

Education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal education	28	7.0	7.0	7.0
	Primary	139	34.8	34.8	41.8
	Secondary	162	40.5	40.5	82.3
	Tertiary	71	17.8	17.8	100.0
	Total	400	100.0	100.0	

4.2 Knowledge and Practice of Open Defecation

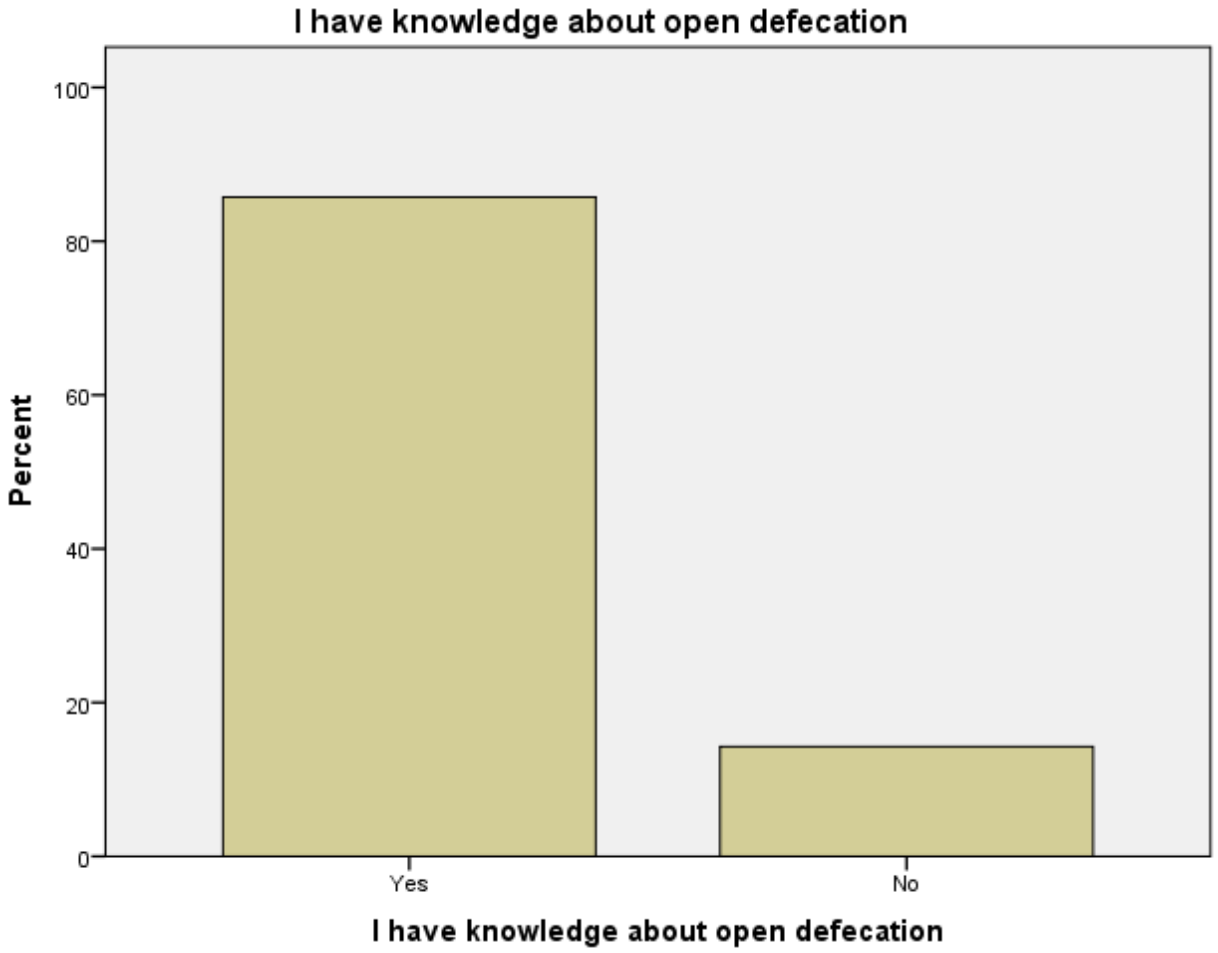
This section presents findings on respondents' awareness, attitudes, and behaviors regarding open defecation. Understanding the level of knowledge and the extent of open defecation practices is crucial in identifying public health risks and designing effective interventions. The results in Tables 4.4 and 4.5 highlight both the knowledge levels and the actual sanitation practices of the respondents.

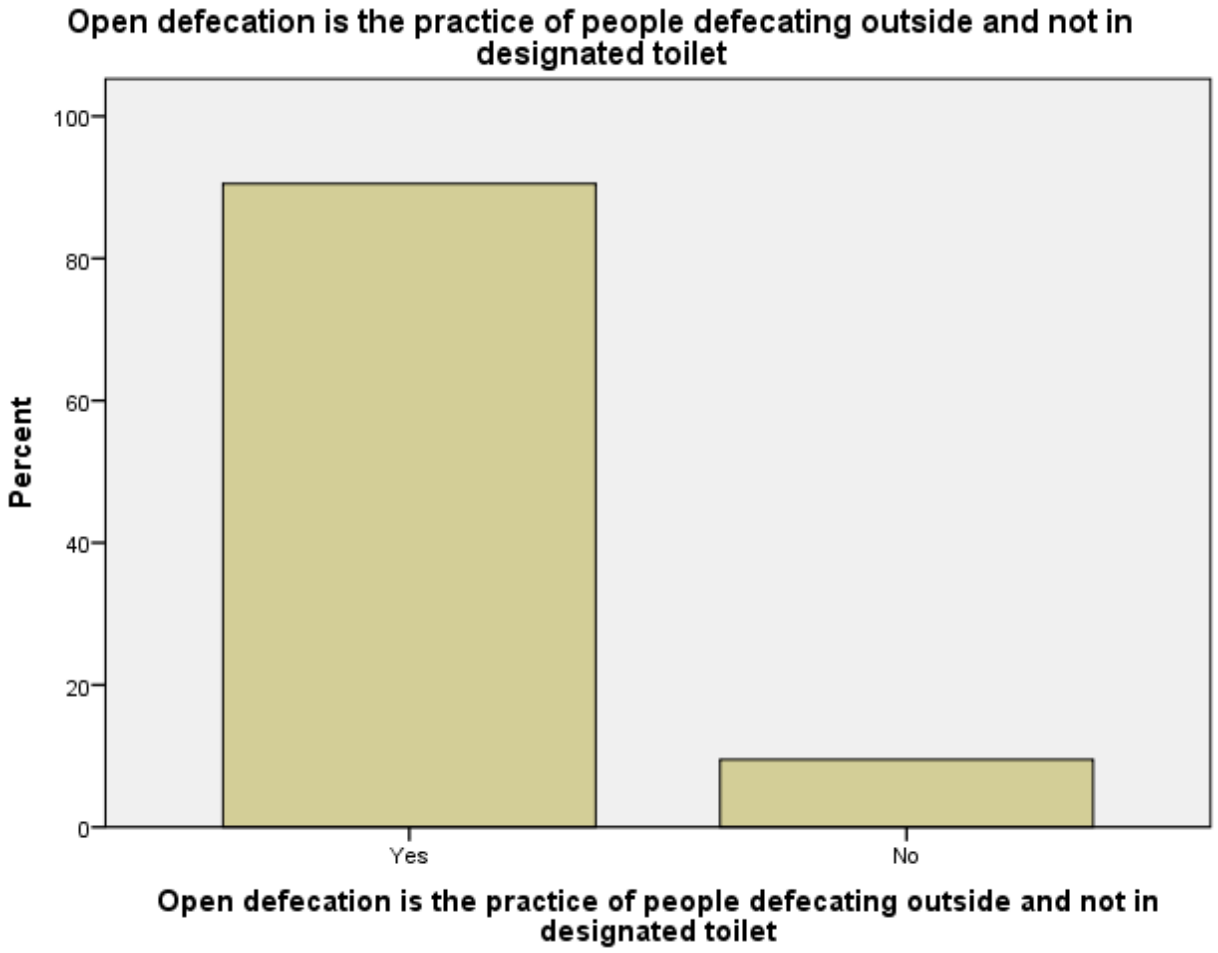
Table 4.4 shows respondents' awareness and understanding of open defecation. The findings reveal that 85.8% reported having knowledge about open defecation, and 90.5% correctly identified it as the act of defecating outside a designated toilet. Furthermore, 92.8% recognized its harmful effects on human health, and 81.0% believed the practice could be avoided. These results suggest a high level of awareness of open defecation and its associated risks, aligning with Objective 1: To assess the prevalence of open defecation in rural communities of Odogbolu Local Government.

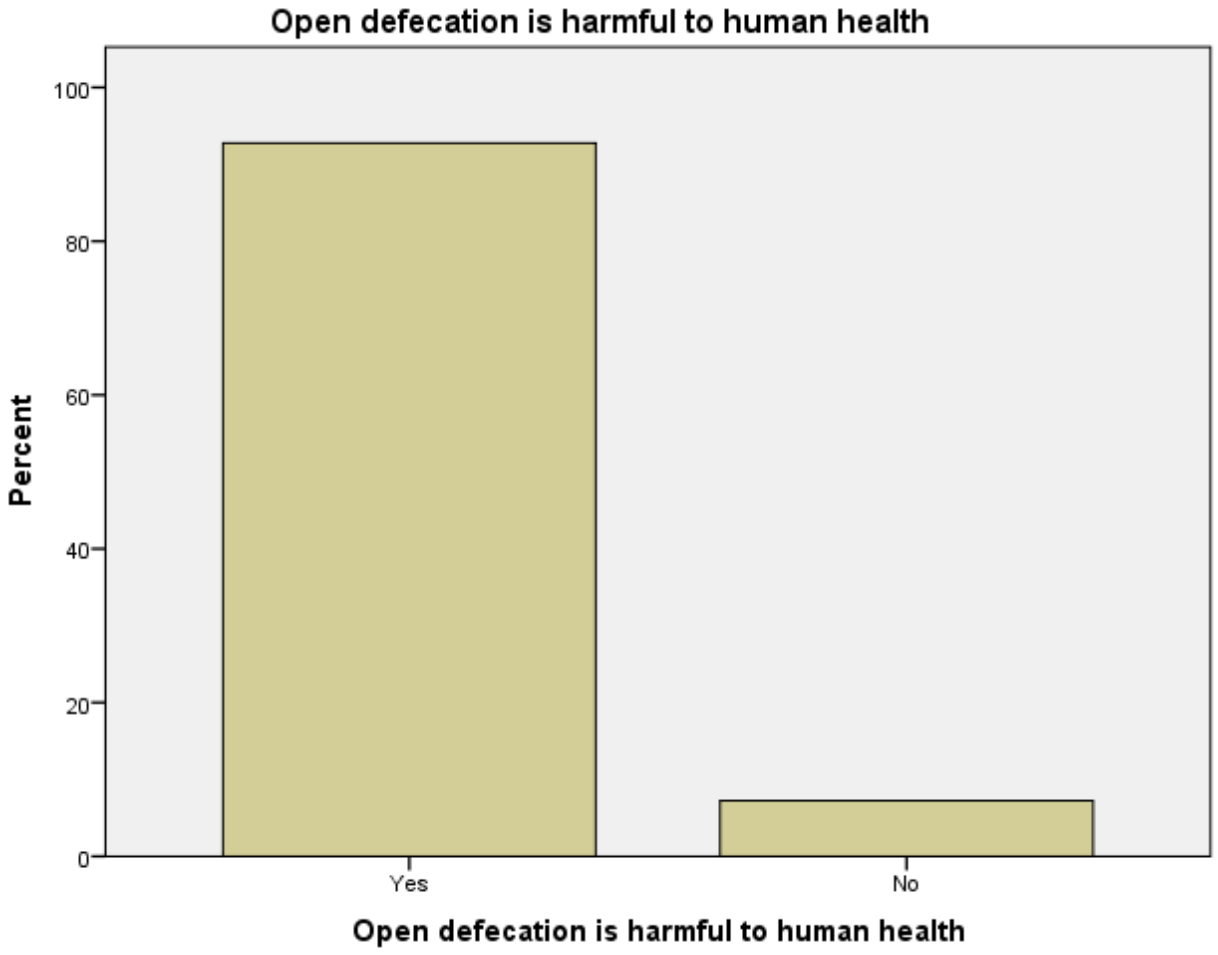
Table 4.5 presents the actual practices related to open defecation. Despite the high awareness reported in Table 4.4, 71.0% of respondents admitted to defecating in the bush, with 59.0% doing so always and 41.0% sometimes. Although 65.3% indicated they had access to a latrine, only 28.2% consistently used a toilet. Hand hygiene after defecation was also inadequate, with only 46.3% reporting that they always washed their hands. This gap between knowledge and behavior highlights persistent cultural habits, inadequate infrastructure usage, and poor hygiene practices, reinforcing Objective 3: To examine the relationship between poor sanitation practices and the spread of parasitic disease.

Table 4.4: Awareness and Understanding of Open Defecation

Variable	Response	Frequency	Percentage (%)
I have knowledge about open defecation	Yes	343	85.8
	No	57	14.2
Open defecation is the act of defecating outside a designated toilet	Yes	362	90.5
	No	38	9.5
Open defecation is harmful to human health	Yes	371	92.8
	No	29	7.2
Practice of open defecation can be avoided	Yes	324	81.0
	No	76	19.0


I have knowledge about open defecation
Percent
100
80
60
40
20
0
Yes
No
I have knowledge about open defecation


Open defecation is the practice of people defecating outside and not in designated toilet
Percent
100
80
60
40
20
0
Yes
No
Open defecation is the practice of people defecating outside and not in designated toilet


Open defecation is harmful to human health
Percent
100
80
60
40
20
0
Yes
No
Open defecation is harmful to human health

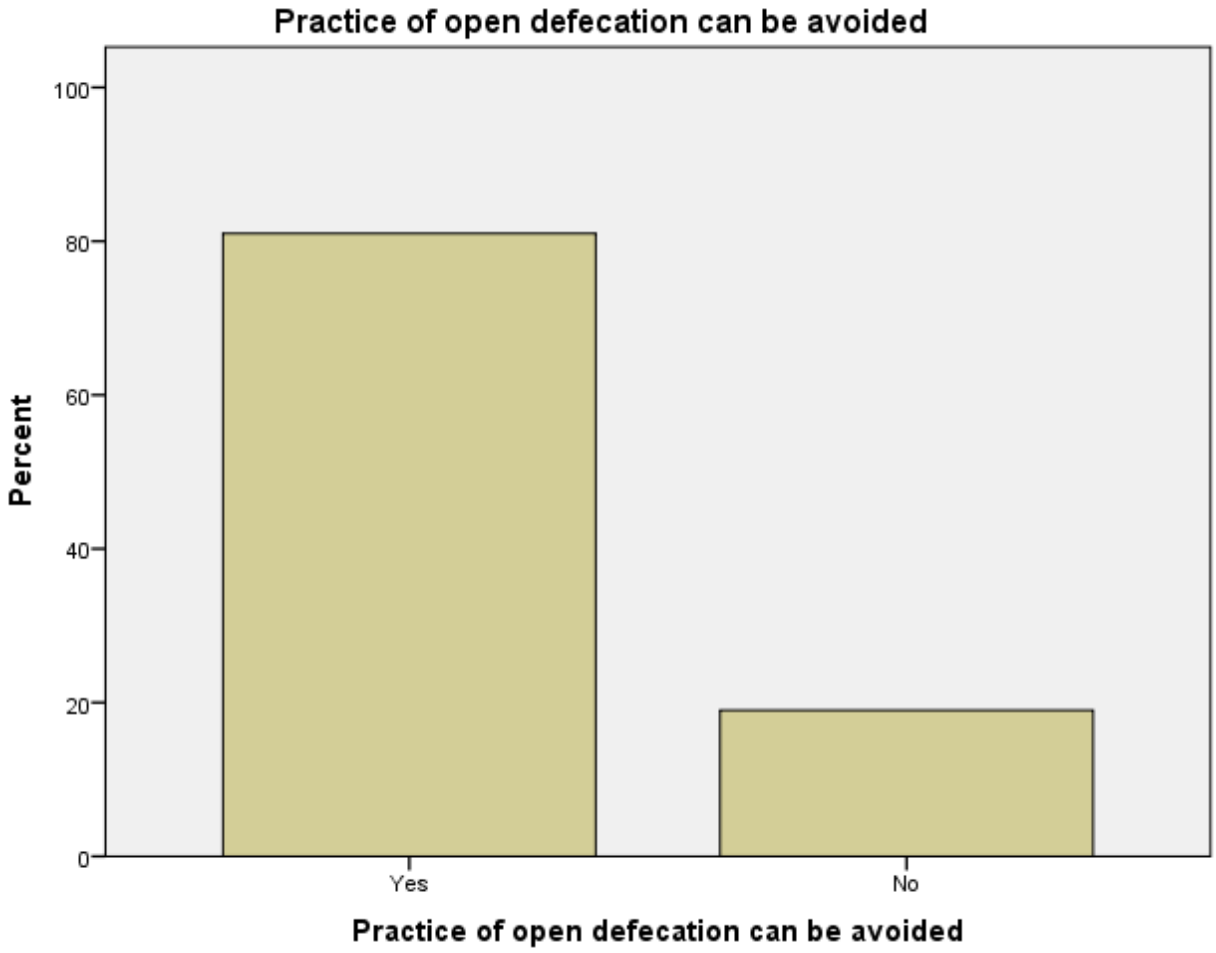
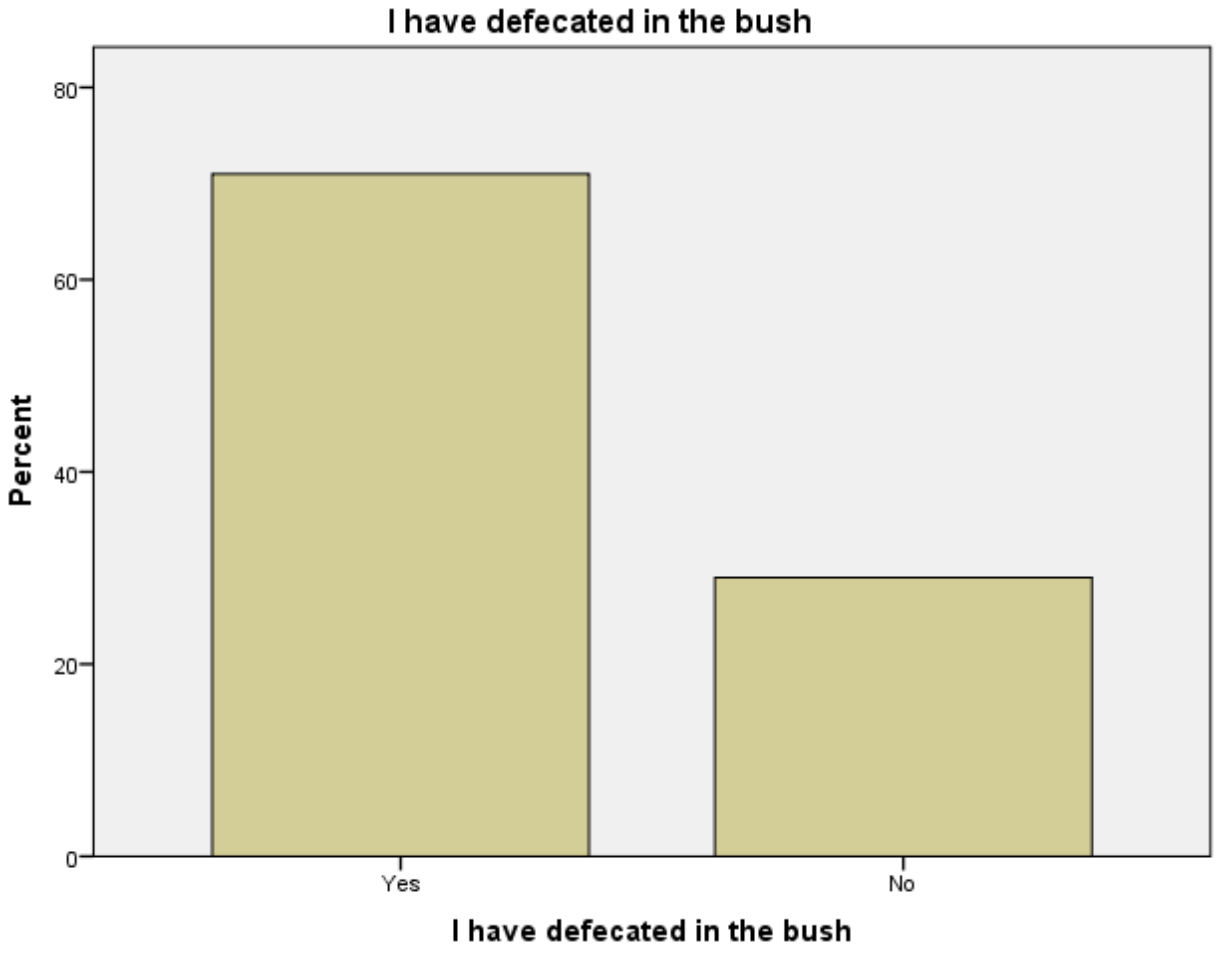
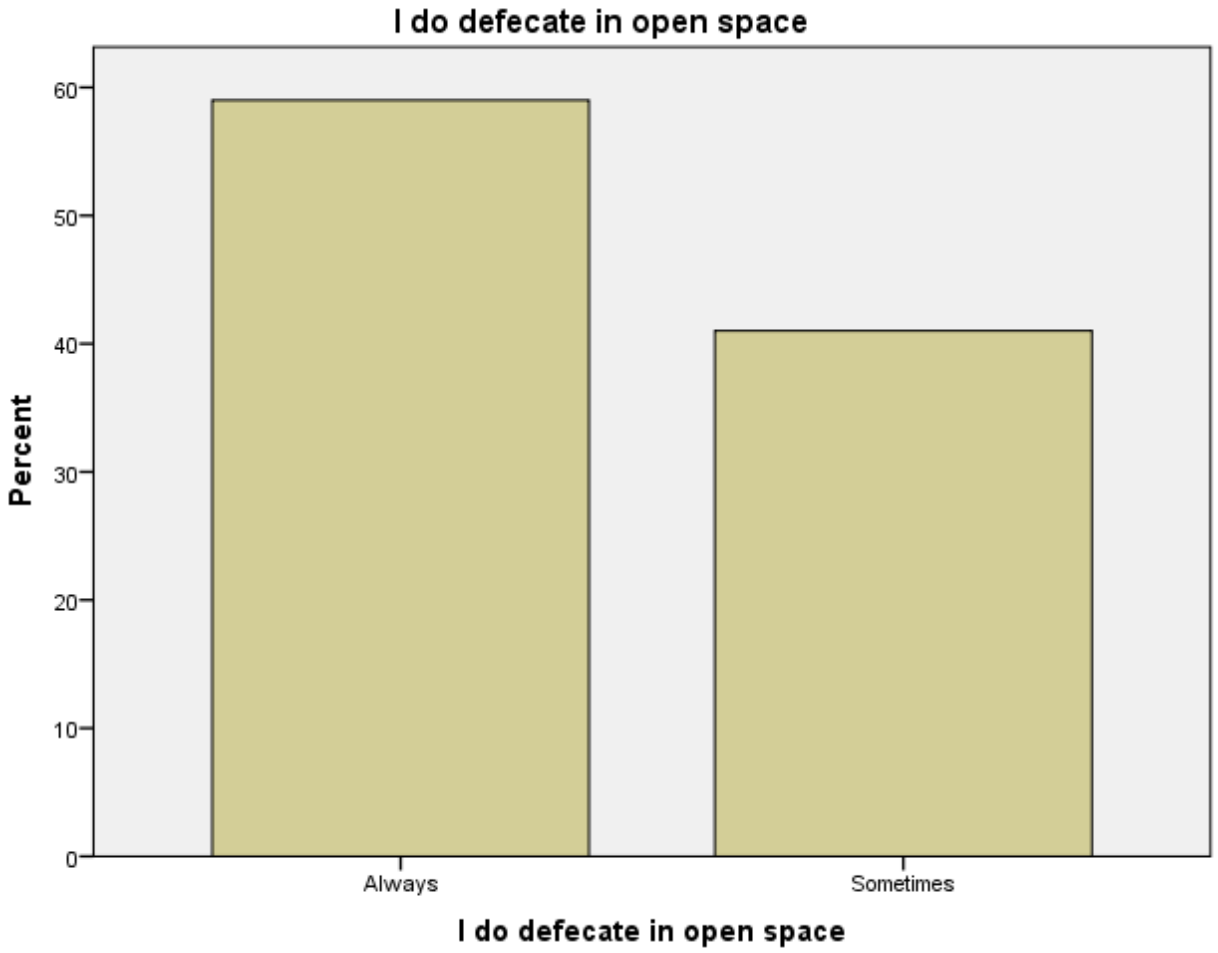
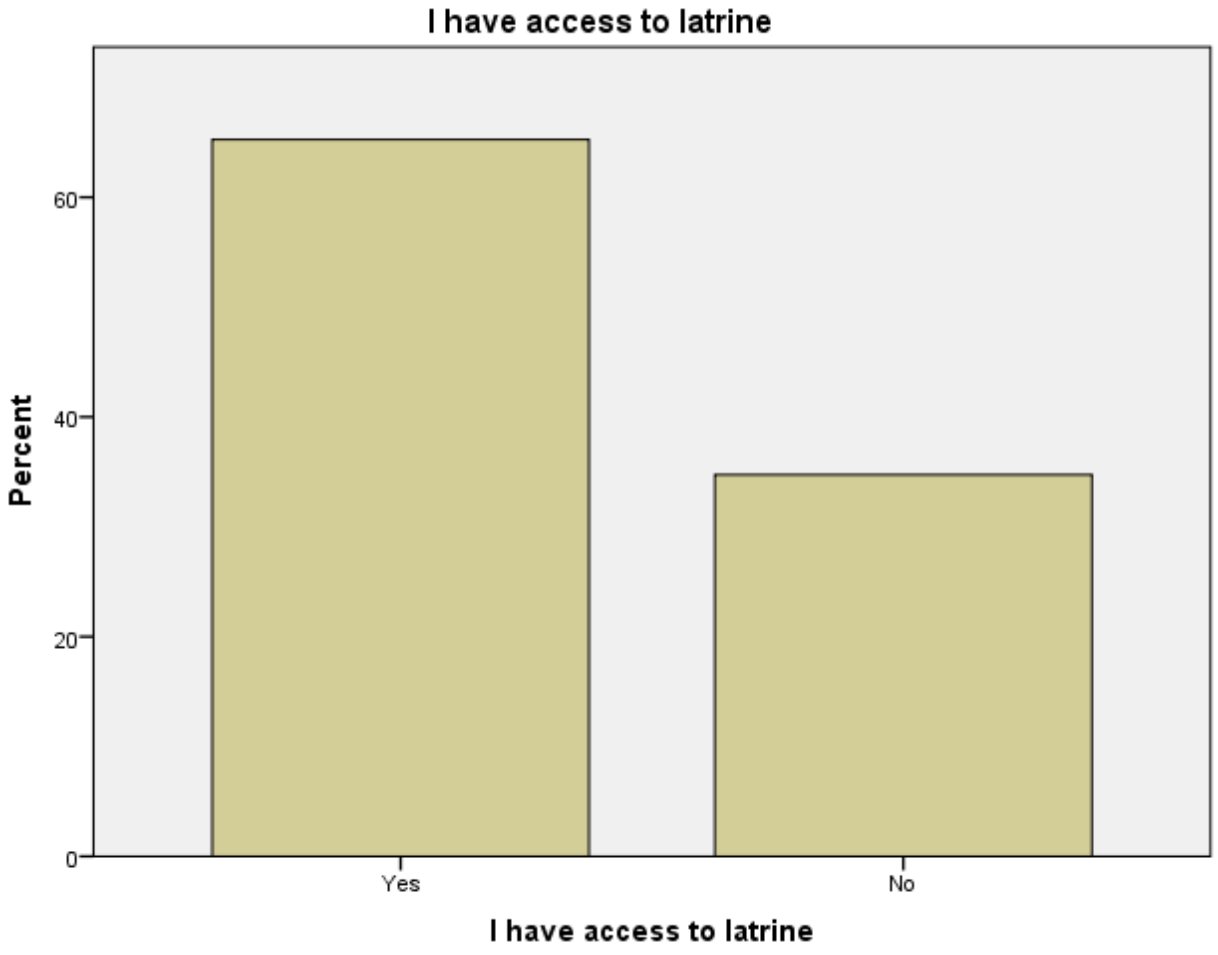

Practice of open defecation can be avoided
Percent
100
80
60
40
20
0
Yes
No
Practice of open defecation can be avoided

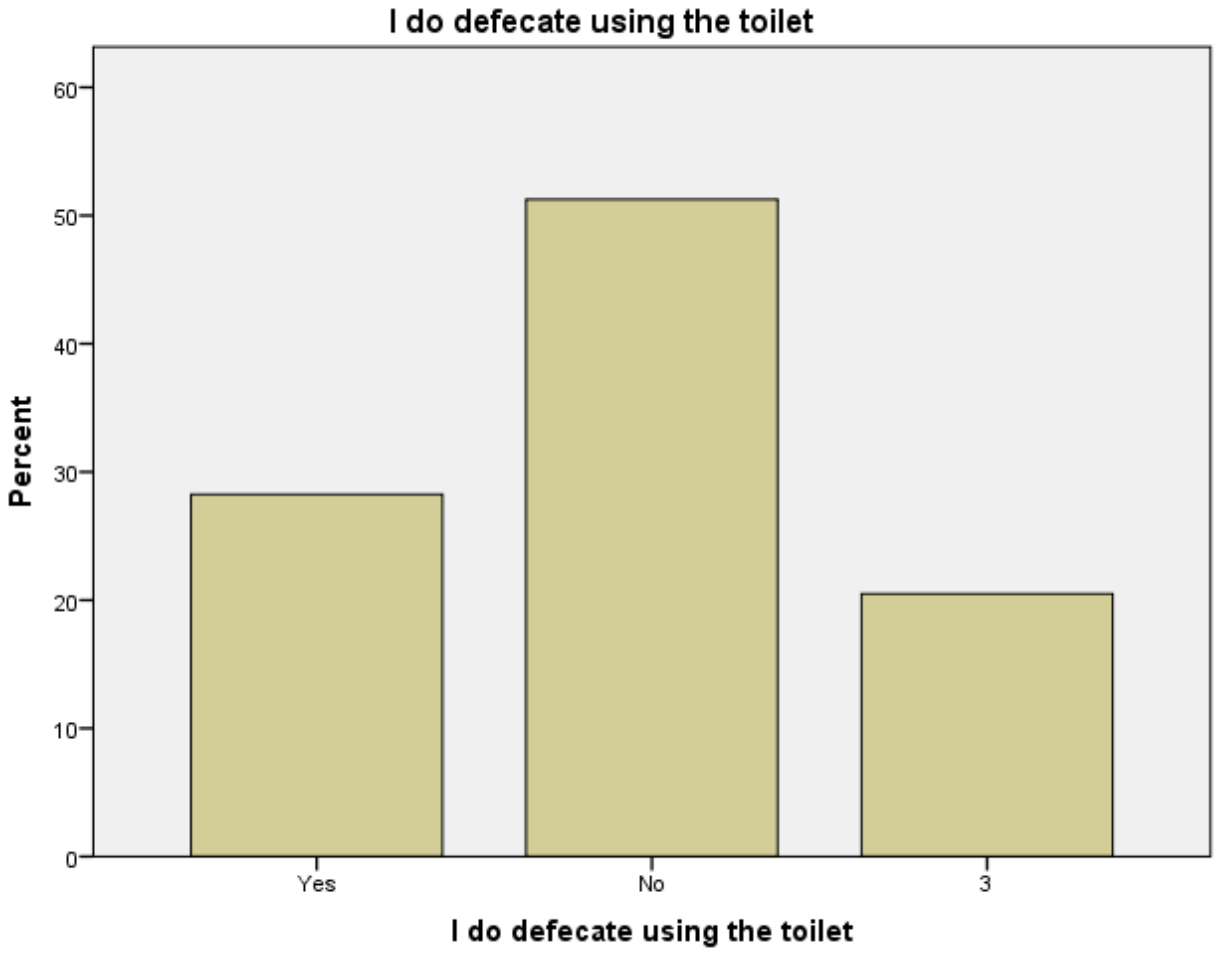
Table 4.5: Practice of Open Defecation

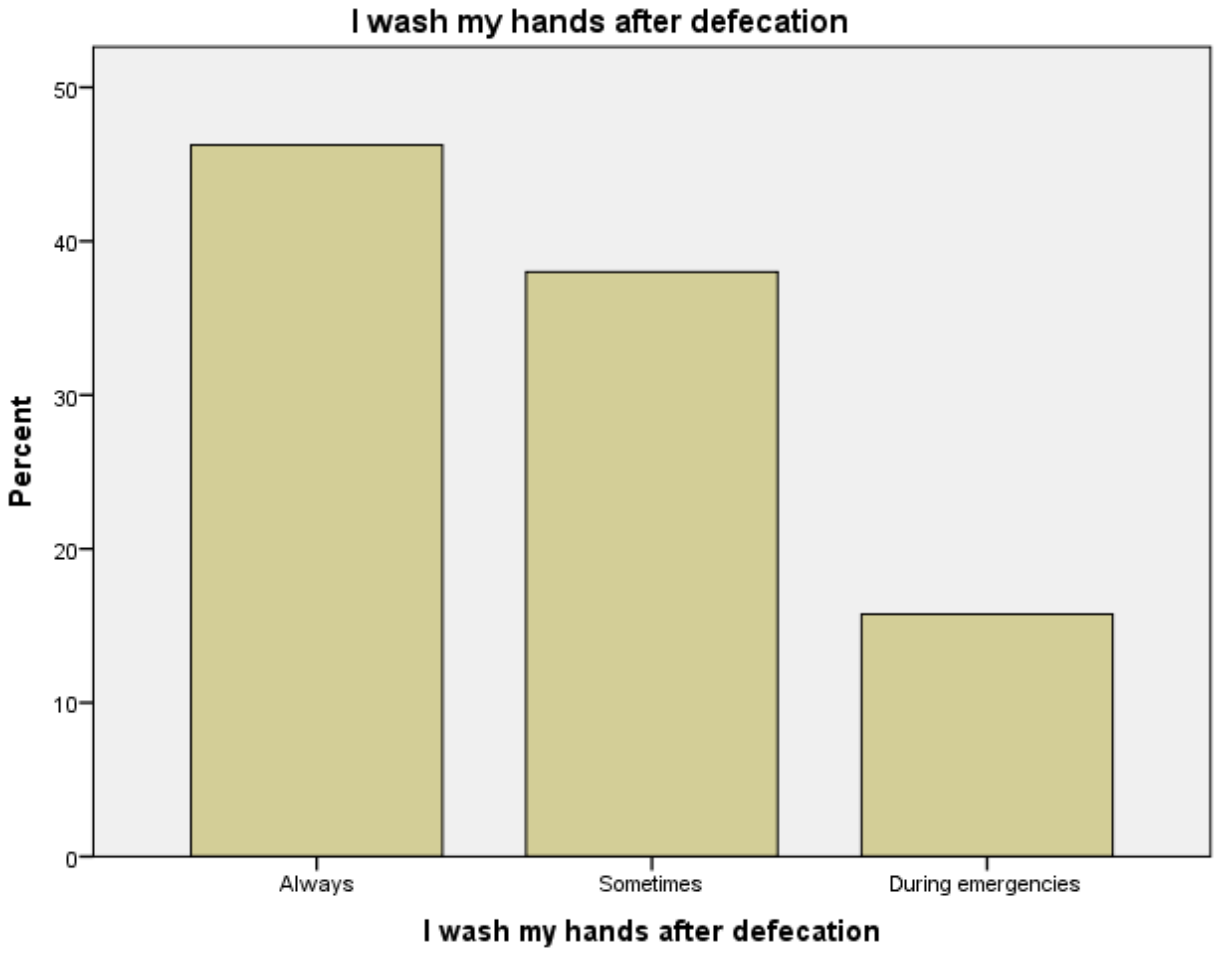
Variable	Response	Frequency	Percentage (%)
I have defecated in the bush	Yes	284	71.0
	No	116	29.0
I defecate in open space	Always	236	59.0
	Sometimes	164	41.0
I have access to latrine	Yes	261	65.3
	No	139	34.8
I use a toilet for defecation	Yes	113	28.2
	No	205	51.2
	No response/Other	82	20.5
I wash my hands after defecation	Always	185	46.3
	Sometimes	152	38.0
	During emergencies	63	15.8


I have defecated in the bush
Percent
80
60
40
20
0
Yes
No
I have defecated in the bush


I do defecate in open space
Percent
60
50
40
30
20
10
0
Always
Sometimes
I do defecate in open space


I have access to latrine
Percent
60
40
20
0
Yes
No
I have access to latrine


I do defecate using the toilet
Percent
60
50
40
30
20
10
0
Yes
No
3
I do defecate using the toilet


I wash my hands after defecation
Percent
50
40
30
20
10
0
Always
Sometimes
During emergencies
I wash my hands after defecation

4.3 Perception of Respondents about Open Defecation

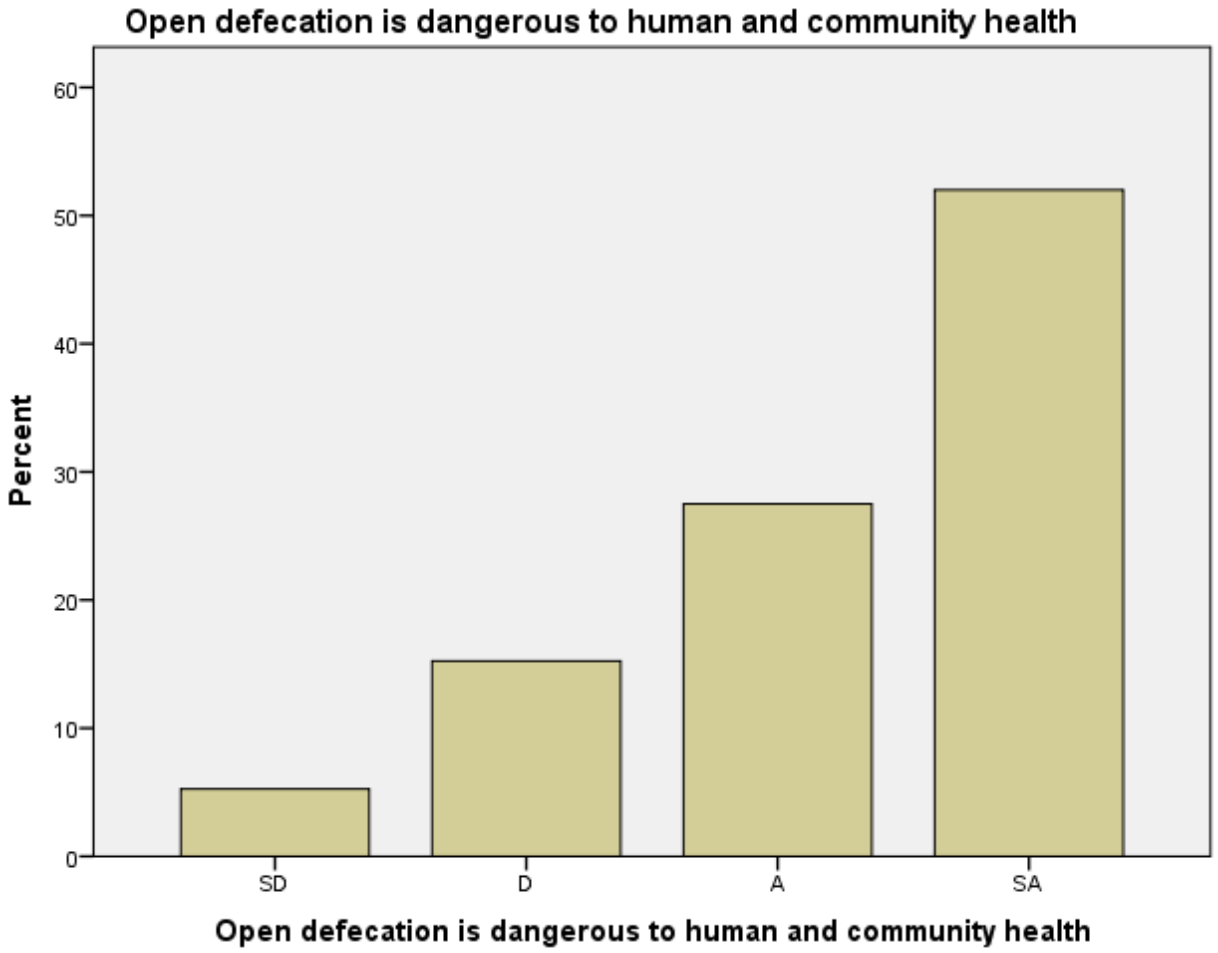
This section presents respondents' attitudes and beliefs toward open defecation, as shown in Table 4.6. A large proportion (79.5%) agreed or strongly agreed that open defecation is dangerous to both human and community health, while over 80% acknowledged that it contributes to environmental degradation. These results suggest that awareness of the negative impacts of open defecation is relatively high among respondents, providing a solid foundation for sanitation improvement programs.

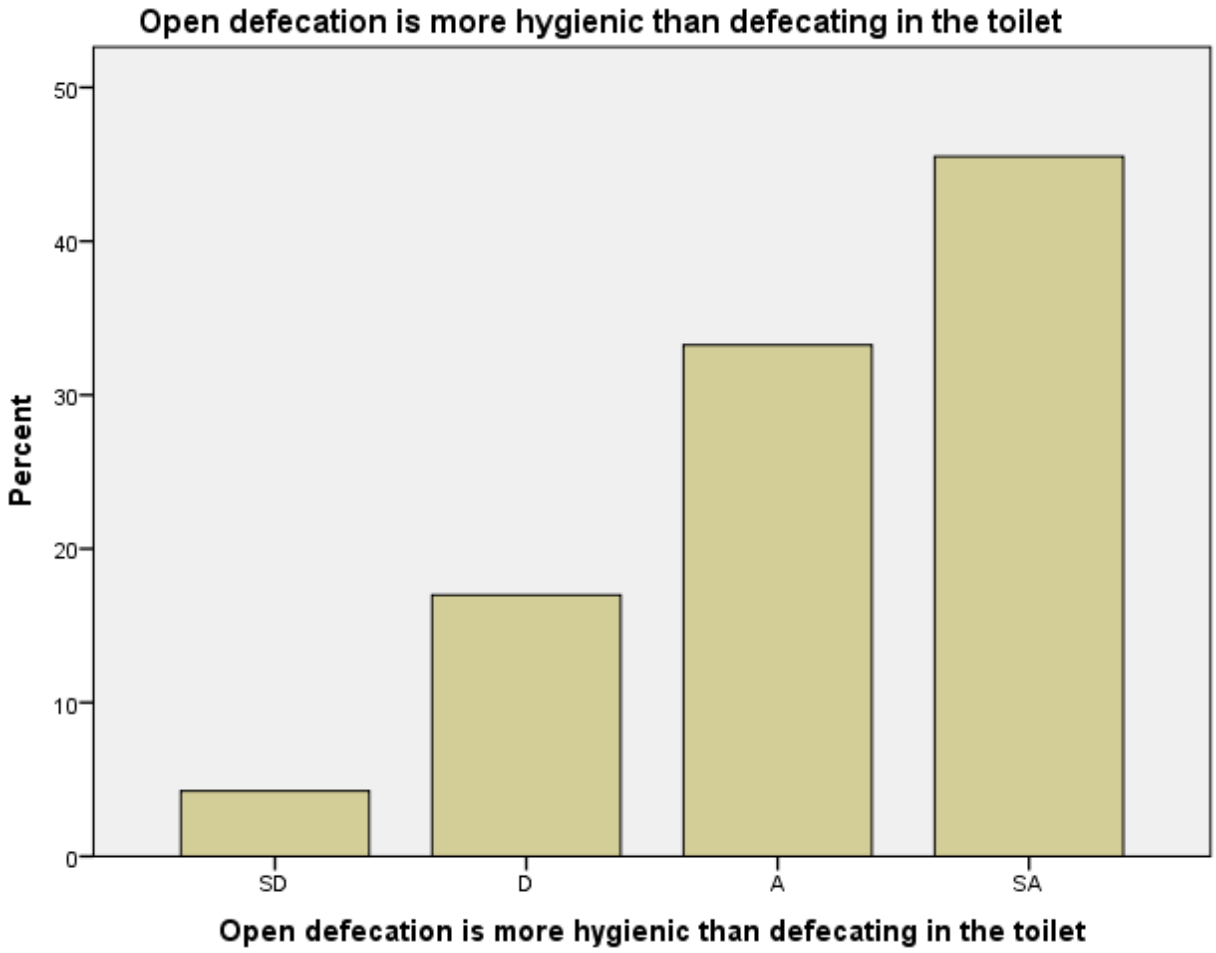
Table 4.6 also highlights some conflicting perceptions. Notably, 78.8% believed that open defecation is more hygienic than using a toilet, and a similar percentage felt that infections can be easily contracted from public toilets. Such views point to mistrust or dissatisfaction with the cleanliness of available public toilet facilities, which may discourage their use even when accessible.

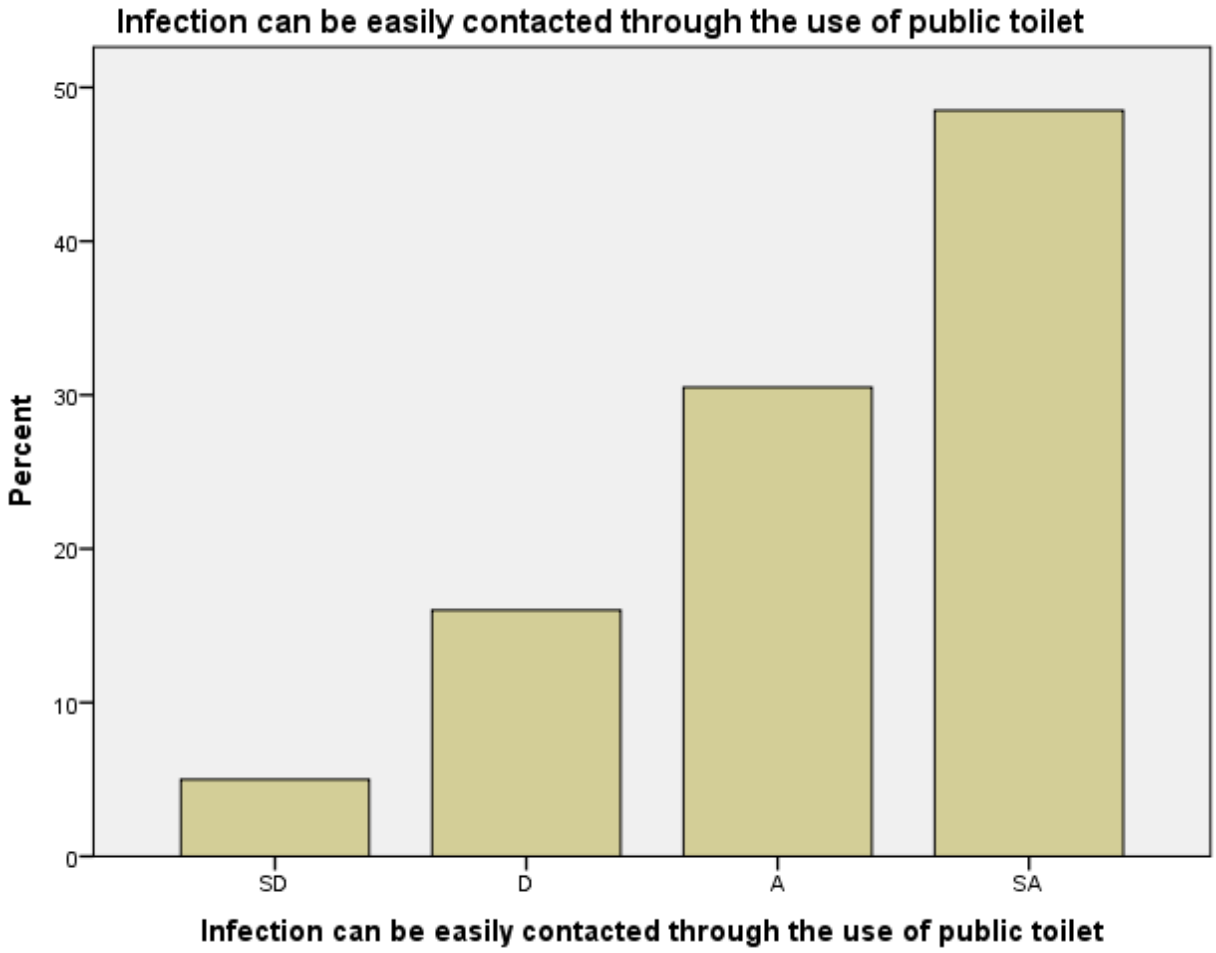
Furthermore, 80.3% of respondents agreed that defecating on farmlands improves soil fertility and benefits the environment. While this belief reflects cultural and traditional practices, it poses health risks, including the transmission of parasites through soil and crops. These findings indicate that any behavior change campaign must address not only awareness but also entrenched cultural beliefs and the quality of sanitation infrastructure.

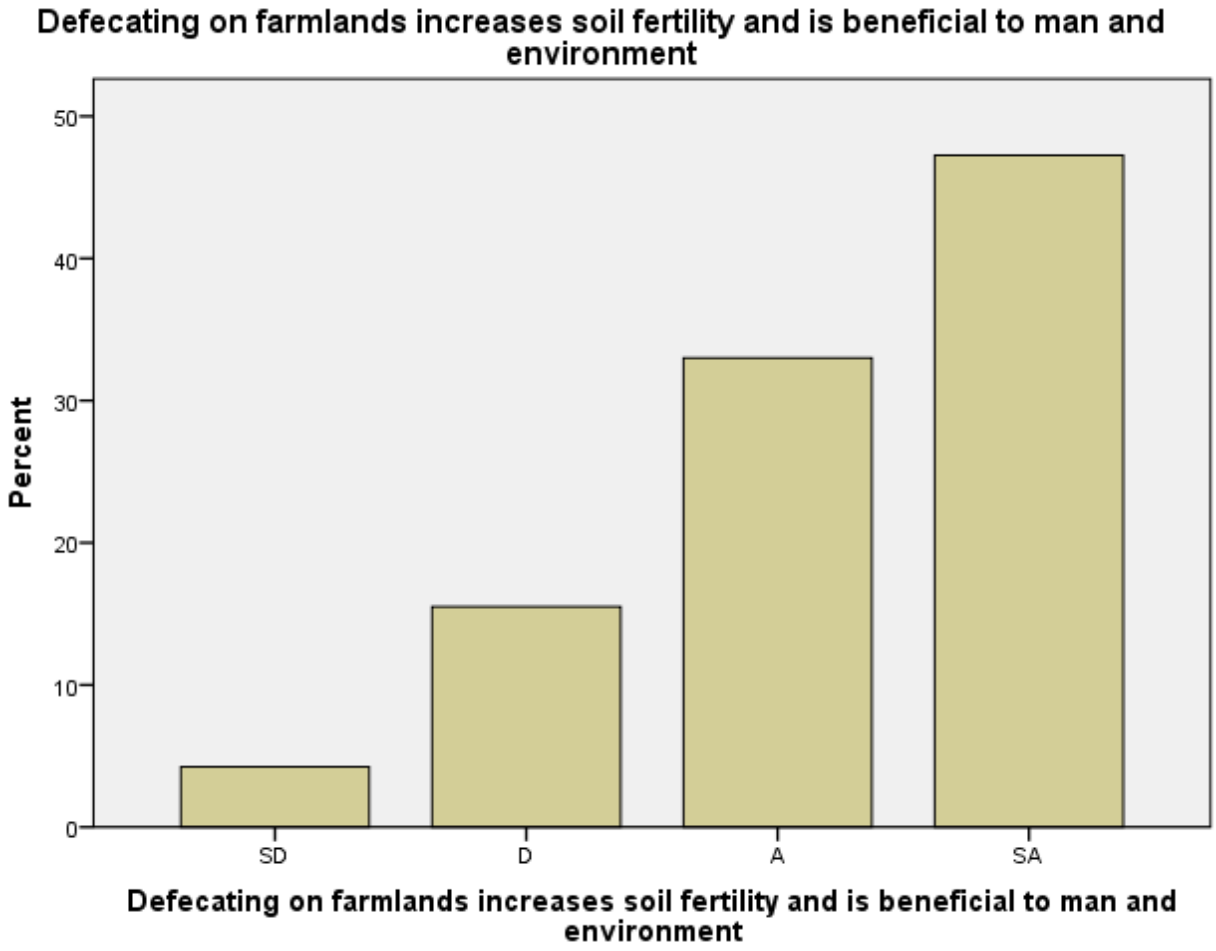
Table 4.6: Respondents' Perception of Open Defecation

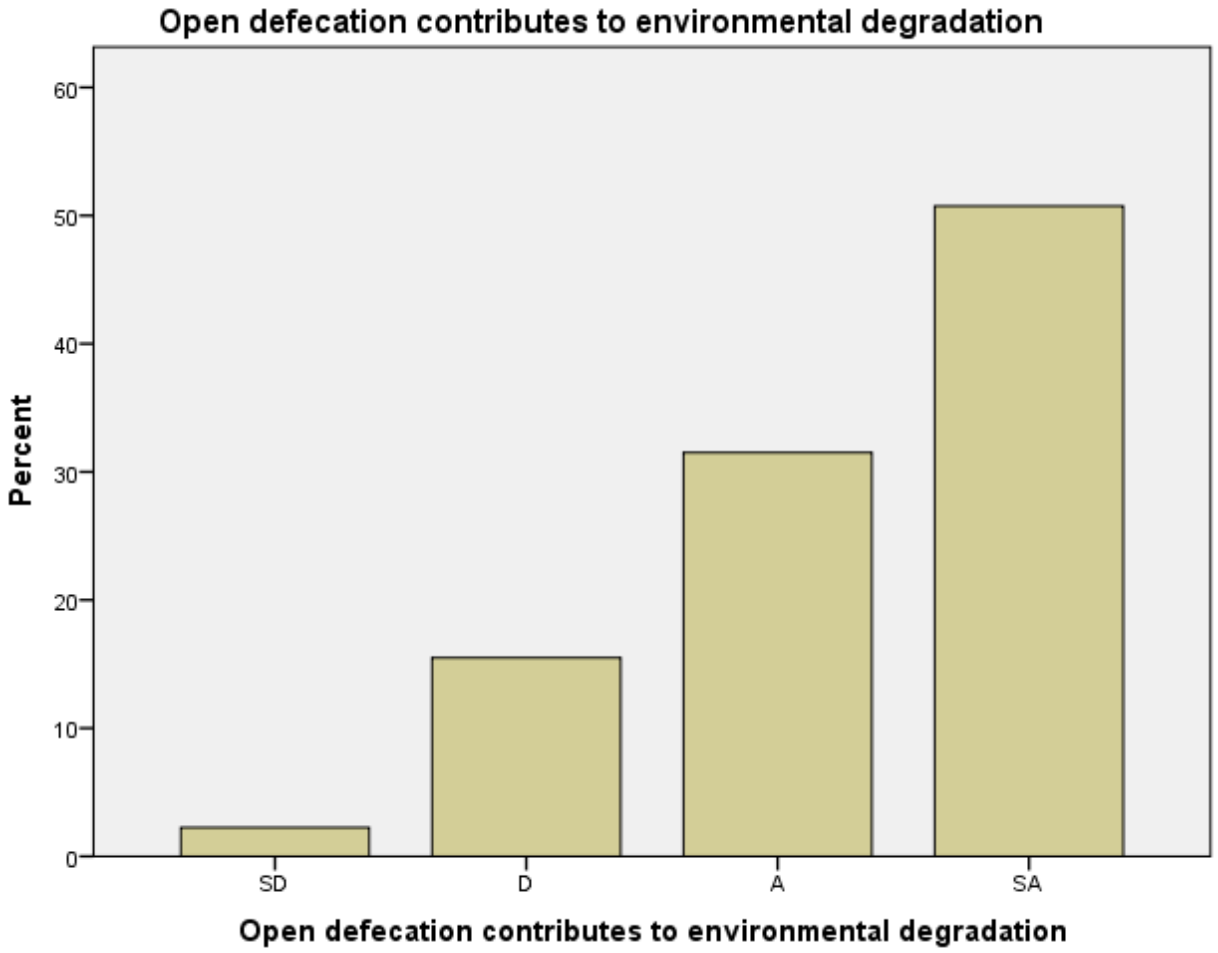
Statement	SD	D	A	SA	Total (%)
Open defecation is dangerous to human and community health	5.3	15.3	27.5	52.0	100
Open defecation is more hygienic than using a toilet	4.3	17.0	33.3	45.5	100
Infection can be easily contracted through the use of public toilets	5.0	16.0	30.5	48.5	100
Defecating on farmlands increases soil fertility and benefits the environment	4.3	15.5	33.0	47.3	100
Open defecation contributes to environmental degradation	2.3	15.5	31.5	50.7	100


Open defecation is dangerous to human and community health
Percent
60
50
40
30
20
10
0
SD
D
A
SA
Open defecation is dangerous to human and community health


Open defecation is more hygienic than defecating in the toilet
Percent
50
40
30
20
10
0
SD
D
A
SA
Open defecation is more hygienic than defecating in the toilet


Infection can be easily contacted through the use of public toilet
Percent
50
40
30
20
10
0
SD
D
A
SA
Infection can be easily contacted through the use of public toilet


Defecating on farmlands increases soil fertility and is beneficial to man and environment
Percent
50
40
30
20
10
0
SD
D
A
SA
Defecating on farmlands increases soil fertility and is beneficial to man and environment


Open defecation contributes to environmental degradation
Percent
60
50
40
30
20
10
0
SD
D
A
SA
Open defecation contributes to environmental degradation

4.4 Health Outcomes and Parasitic Infections

This section presents the health consequences of open defecation as reported by respondents, including symptoms, medical history, and laboratory-confirmed parasitic infections. The results address Objective 2 (to identify common parasitic infections associated with open defecation) and Objective 3 (to examine the relationship between poor sanitation practices and the spread of parasitic disease).

Table 4.7 shows the distribution of self-reported symptoms of parasitic infections in the past three months. Fatigue or weakness (36.3%) and diarrhea (33.3%) were the most commonly reported symptoms, both typical indicators of parasitic infection. Soil-transmitted helminths and protozoal infections were also reported by over a quarter of respondents, suggesting a significant parasitic disease burden in the study area. These findings reflect the health risks posed by continued exposure to open defecation environments.

Table 4.8 presents respondents' treatment and diagnosis history. Only 32.8% sought medical care for their symptoms, which may indicate limited healthcare access or poor health-seeking behavior. A total of 20.5% had been previously diagnosed with a parasitic infection, which aligns with the symptom patterns observed. Additionally, 29.3% had used antibiotics recently, which could suggest self-treatment or prior consultations, though not necessarily targeted against parasitic infections.

Table 4.9 summarizes the frequency of parasitic diagnoses among respondents. A striking 81% had been diagnosed at least once, with 21.8% diagnosed four or more times. These high recurrence rates point to persistent reinfection, likely due to repeated exposure to contaminated environments

and ongoing unsafe defecation practices. This pattern underscores the cyclical nature of infection in settings where sanitation is inadequate.

Table 4.10 details the laboratory findings of parasitic infections among respondents. More than half (54.8%) tested positive for at least one parasite, with Entamoeba spp. being the most prevalent (19.8%), followed by Hookworm (17.8%) and Schistosoma spp. (17.3%). These parasites are strongly linked to poor sanitation and open defecation, providing concrete biological evidence that such practices are a major driver of parasitic disease transmission in the area.

Table 4.7: Reported Symptoms of Parasitic Infection (Past 3 Months)

Symptom	Response	Frequency	Percentage (%)
Soil-Transmitted Helminths	Yes	113	28.2
Protozoal Infection	Yes	108	27.0
Schistosomiasis	Yes	41	10.3
Nausea and Vomiting	Yes	81	20.3
Diarrhea	Yes	133	33.3
Fatigue or Weakness	Yes	145	36.3

Table 4.8: Treatment and Diagnosis History

Variable	Response	Frequency	Percentage (%)
Sought treatment for symptoms	Yes	131	32.8
Previously diagnosed with parasitic infection	Yes	82	20.5
Used antibiotics recently	Yes	117	29.3
Known underlying health condition	Yes	43	10.8

Table 4.9: Frequency of Parasitic Diagnosis

Times Diagnosed	Frequency	Percentage (%)
None	76	19.0
Once	79	19.8
Twice	83	20.8
Thrice	75	18.8
Four or more	87	21.8

Table 4.10: Identified Parasites from Laboratory Findings

Parasite Identified	Frequency	Percentage (%)
Entamoeba	79	19.8
Hookworm	71	17.8
Schistosoma	69	17.3
None	181	45.3

4.5 Discussion of Findings

This section discusses the key findings of the study in relation to the research objectives. It also connects the results to existing literature, highlighting trends, confirming previous knowledge, or identifying unique observations within the Odogbolu Local Government Area.

Prevalence of Open Defecation

The study revealed that open defecation is still widely practiced in rural communities of Odogbolu. Despite high awareness of its dangers, 71% of respondents admitted to defecating in the bush, and 59% said they do so regularly. These findings confirm the continued prevalence of open defecation in rural Nigeria, as previously reported by Abdullahi et al. (2023), who noted widespread open defecation due to poor infrastructure and traditional habits. The gap between knowledge and behavior indicates that awareness alone is not sufficient to drive behavioral change without addressing access and cultural factors.

Parasitic Infections Associated with Open Defecation

The study identified several parasitic infections among respondents, with *Entamoeba spp.* (19.8%), *Hookworm* (17.8%), and *Schistosoma spp.* (17.3%) being the most common. Symptoms such as diarrhea (33.3%), fatigue (36.3%), and nausea were also frequently reported. These results are consistent with findings by Ekhoragbon (2024), who emphasized the link between poor sanitation and parasitic disease transmission in rural Nigeria. The high rate of infections, including laboratory-confirmed cases, demonstrates the direct health risks of open defecation, fulfilling the second objective of this study.

Sanitation Practices and Health Outcomes

While 65.3% of respondents claimed to have access to a latrine, only 28.2% regularly used one. A majority still rely on open spaces, citing fear of public toilet hygiene or cultural beliefs. This indicates a disconnect between facility availability and utilization, which supports the findings of Saleem et al. (2019) on the role of fear, privacy, and cleanliness in toilet avoidance. Moreover, a large proportion of respondents believed that using public toilets leads to infections, reinforcing the idea that poor maintenance deters proper sanitation practices.

Community Perception and Cultural Beliefs

The perception section showed that while respondents recognized the health and environmental dangers of open defecation, many also believed it could be beneficial — especially on farmlands. These cultural beliefs present a significant barrier to change. Without addressing the underlying traditional views and improving the condition of toilet facilities, efforts to eliminate open defecation may not succeed. This finding aligns with WHO and UNICEF reports that emphasize the need for community engagement and infrastructure development in tackling open defecation globally.

In summary, the study demonstrates that open defecation remains a deeply rooted practice influenced by a combination of limited infrastructure, poor hygiene standards, cultural beliefs, and health misconceptions. While awareness exists, sustainable change will require targeted interventions that improve facilities, correct misinformation, and promote behavioral shifts.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

This study assessed the prevalence of open defecation and its role in the spread of parasitic infections in rural communities of Odogbolu Local Government Area. A total of 400 respondents participated, providing insights into their sanitation practices, health symptoms, and cultural perceptions.

The findings revealed that while awareness of the dangers of open defecation is high (over 90% acknowledged its risks), the practice remains prevalent. About 71% of respondents admitted to defecating in bushes, and only 28.2% consistently used toilet facilities. Many respondents believed that public toilets are unhygienic and that open defecation is more sanitary — highlighting a major behavioral and infrastructural gap.

Health-wise, common symptoms such as diarrhea (33.3%), fatigue (36.3%), and parasitic infections such as *Entamoeba*, *Hookworm*, and *Schistosoma* were recorded. Cultural beliefs, such as the idea that human waste improves soil fertility, further reinforced the continuation of open defecation despite known risks.

These findings confirm that open defecation is both a health and a behavioral issue, driven by poor sanitation infrastructure, misinformation, and deep-rooted traditional practices.

5.2 Conclusion

Open defecation continues to pose a major public health threat in Odogbolu's rural communities. While most residents are aware of its dangers, practical limitations and cultural attitudes hinder the adoption of safer sanitation practices. The study clearly establishes a relationship between poor sanitation and the prevalence of parasitic infections, as evidenced by both reported symptoms and laboratory diagnoses.

Addressing this issue requires more than raising awareness — it demands improvements in toilet infrastructure, health education, and culturally appropriate community engagement. Without a holistic intervention, the cycle of open defecation and disease transmission is likely to persist.

5.3 Recommendations

Based on the study's findings, the following recommendations are made:

1. **Improve Sanitation Infrastructure:** Local government and health agencies should prioritize the construction and maintenance of hygienic, accessible, and affordable toilet facilities in rural areas.

2. **Promote Health Education:** Awareness campaigns should go beyond general information and directly address local myths — such as the belief that open defecation is more hygienic or beneficial to soil.

3. **Community Engagement and Participation:** Involve community leaders, farmers, and residents in sanitation programs to ensure ownership, accountability, and cultural alignment.

4. **Public Toilet Management:** Where public toilets exist, routine cleaning and security should be ensured to restore public trust in their use.

5. **School and Youth Education Programs:** Integrate sanitation and hygiene education into school curricula to shape long-term attitudes and behaviors.

6. **Regular Health Screening and Deworming:** Organize community-based health checks, especially in high-risk areas, to treat infections early and monitor improvements.

5.4 Suggestions for Further Studies

Future research could explore:

- The psychological and behavioral factors that influence defecation habits in rural populations.
- Comparative studies between communities with and without improved sanitation.
- Longitudinal studies to assess the long-term impact of interventions on open defecation and public health outcomes.